

Yes. I would turn the challenge into a **small digital wallet + reliable money movement system**, rather than trying to build a huge banking app.

Here is the feature list I would give your team as the **implementation roadmap**, in priority order.

Complete Feature List — Serially

Phase 1 — User & Account

1. User Registration

- Name
- Email/phone/username
- Password
- Create account automatically
- Initial balance: **BDT 100,000**

2. Initial Account Funding

- Automatically credit BDT 100,000 at registration.
- Record it as an INITIAL_CREDIT ledger transaction.
- Never silently create money without a record.

3. Login & Authentication

- Secure password hashing
- JWT authentication
- Token expiration
- Logout/session handling

4. User Profile

- Name
- Username/phone/email
- Account status
- Account creation date

Phase 2 — Wallet / Dashboard

5. Wallet Dashboard

Show:

Available Balance

BDT 97,500

[Send Money]

[Request Money]

Recent Transactions

6. Balance Management

- Current balance
- Available balance
- Account status
- Never trust balance information supplied by the frontend.

Phase 3 — Send Money

7. Search User

Search by:

- Username
- Email
- Phone

Example:

Search: Bob

Bob Rahman
bob@example.com

[Send Money]

8. Send Money

User enters:

Receiver: Bob

Amount: BDT 2,500

Note: Lunch

Then:

[Confirm Transfer]

9. Transfer Validation

Reject:

- Amount ≤ 0
- Amount above available balance
- Non-existent receiver
- Suspended account
- Self-transfer
- Invalid amount format

10. Transaction Confirmation

Before sending:

Send BDT 2,500 to Bob?

[Cancel] [Confirm]

This prevents accidental transfers.

Phase 4 — The Most Important Part

This is where your project becomes more than CRUD.

11. Atomic Money Transfer

Transfer must behave as:

BEGIN TRANSACTION

Lock accounts

↓

Check balance

↓

Debit sender

↓

Credit receiver

↓

Create transaction

↓

Create ledger entries

↓

COMMIT

If anything fails:

ROLLBACK EVERYTHING

12. Concurrency Control

Handle:

Alice balance = 100,000

Request A → 70,000

Request B → 50,000

arriving simultaneously.

The system must ensure Alice cannot spend:

120,000

when she only has:

100,000

Use database transaction + row-level locking.

13. Idempotency

Protect against:

User taps Send twice

or:

Request succeeds

↓

Network timeout

↓

Client retries

The same request should **not transfer twice**.

Use an:

`idempotency_key`

with a unique database constraint.

14. Transaction Status

Every transfer should have a clear state:

PENDING

PROCESSING

COMPLETED

FAILED

CANCELLED

For the MVP, you can initially use:

PENDING

COMPLETED

FAILED

15. Database Rollback / Failure Handling

Demonstrate:

Debit succeeds

Credit fails

↓

ROLLBACK

↓

Sender gets money back

No partial transaction.

Phase 5 — Transaction & Ledger

16. Transaction History

Show:

Today

↓ Sent BDT 2,500
To Bob

↑ Received BDT 1,200
From Alice

↓ Sent BDT 500
To Charlie

17. Transaction Details

Click a transaction:

Transaction ID
TX-82931

Status
COMPLETED

Amount
BDT 2,500

From
Alice

To

Bob

Date

29 Aug 2026

18. Double-entry-style Ledger

For every successful transfer:

Alice

DEBIT 2,500

Bob

CREDIT 2,500

This gives you an auditable financial history.

19. Ledger Immutability

Don't allow users to simply edit/delete financial records.

Instead:

Transaction created

↓

Historical record remains

This is important for trust and auditing.

Phase 6 — Request Money

20. Request Money

Alice:

Request BDT 1,200

from Bob

21. Incoming Money Requests

Bob sees:

Alice requested

BDT 1,200

[Pay] [Decline]

22. Accept Request

Bob presses Pay:

Bob → Alice

BDT 1,200

The same **atomic transfer engine** should be used.

Don't create a completely separate payment system.

23. Decline Request

Bob can reject it.

Status:

DECLINED

24. Request Expiration

For example:

Request BDT 1,200

Expires in 24 hours

After expiration:

EXPIRED

Phase 7 — Notifications

25. In-app Notifications

Examples:

You received BDT 2,500 from Bob.

Alice requested BDT 1,200 from you.

Your transfer of BDT 500 was successful.

Your transfer failed due to insufficient balance.

26. Notification Center

Something like:

Notifications

- Received BDT 2,500
- Money request from Alice
- Transfer completed

For the hackathon, **in-app notifications are enough**. Don't waste time on SMS.

Phase 8 — Security

27. Password Hashing

Never store:

```
password = 123456
```

Store a secure hash.

28. JWT Authentication

Every protected API verifies the user's identity.

29. Authorization

A user must not be able to do:

```
GET /accounts/another-user
```

and see someone else's private information.

30. Input Validation

Validate everything server-side.

Especially:

```
amount
receiver
transaction ID
request ID
```

31. Rate Limiting

Protect endpoints such as:

```
POST /login
POST /transfer
POST /request-money
```

from abuse.

For example:

```
Too many requests
      ↓
429 Too Many Requests
```

Phase 9 — Reliability

32. Network Retry Safety

Client can retry safely because of idempotency.

33. Database Constraints

Use:

- Foreign keys
- Unique constraints
- Check constraints
- NOT NULL constraints

For example:

`amount > 0`

should also be enforced at the database level where appropriate.

34. Transaction Audit Trail

Record:

Who

What

When

Amount

From

To

Status

Transaction ID

35. Error Handling

Don't return:

500 Internal Server Error
for everything.

Give meaningful errors:

Insufficient balance

Receiver not found

Transaction already processed

Account suspended

Invalid amount

Phase 10 — Testing

This is **very important** for your specific challenge.

36. Normal Transfer Test

`A = 100,000`

A → B = 2,500

A = 97,500

B = 102,500

37. Insufficient Balance Test

A = 1,000

A → B = 2,000

FAIL

Balance remains:

1,000

38. Concurrent Transfer Test

This should be one of your **main demonstrations**.

A = 100,000

Request 1 → 70,000

Request 2 → 50,000

Send them concurrently.

Expected:

ONE succeeds

ONE fails

Never:

A = -20,000

and never incorrectly allow both.

39. Duplicate Request Test

Send the same request twice:

idempotency_key = ABC123

Expected:

ONE transaction

40. Failure/Rollback Test

Force an error during the transaction.

Expected:

No partial balance update

Phase 11 — Frontend

Since you're using Flutter, keep the UI simple.

41. Splash / App Entry

42. Login Screen

43. Registration Screen

44. Home/Dashboard

45. Send Money Screen

46. Select/Search User Screen

47. Transfer Confirmation Screen

48. Transfer Result Screen

Example:

✓ Transfer Successful

BDT 2,500

Sent to
Bob Rahman

Transaction ID
TX-82931

49. Request Money Screen

50. Pending Requests Screen

51. Transaction History

52. Transaction Details

53. Notifications

54. Profile/Settings

Phase 12 — Engineering Demonstration

I would add these specifically for the judges.

55. Transaction ID / Reference

Every transaction gets something like:

TX-20260829-82931

Users can reference it when reporting an issue.

56. System Health

Optional simple admin/engineering endpoint:

System Status: Healthy

Database: Connected

Not necessary for users, but useful during demo.

57. Basic Observability

Log important events:

TRANSFER_STARTED

TRANSFER_COMPLETED

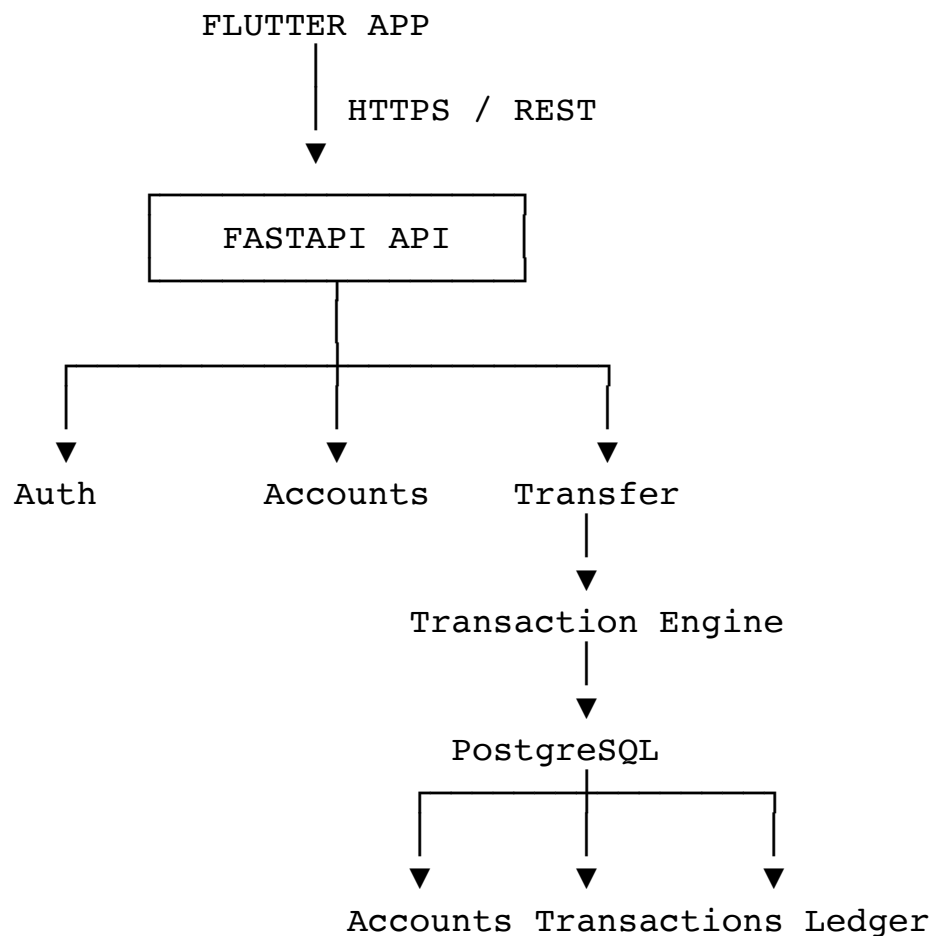
TRANSFER_FAILED

TRANSFER_ROLLED_BACK

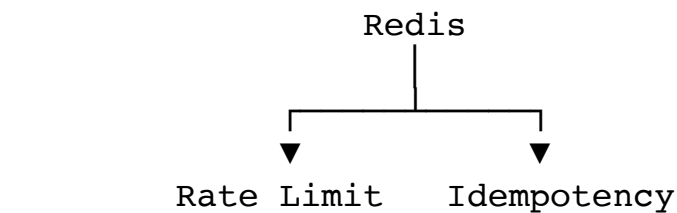
Don't log passwords, JWTs, or sensitive information.

The architecture I'd finally choose

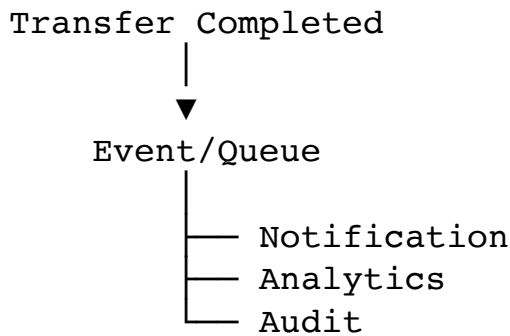
For your team, I'd keep it:



Then optionally:



And later:



But don't start with Kafka/RabbitMQ/microservices. Build the reliable transaction core first.

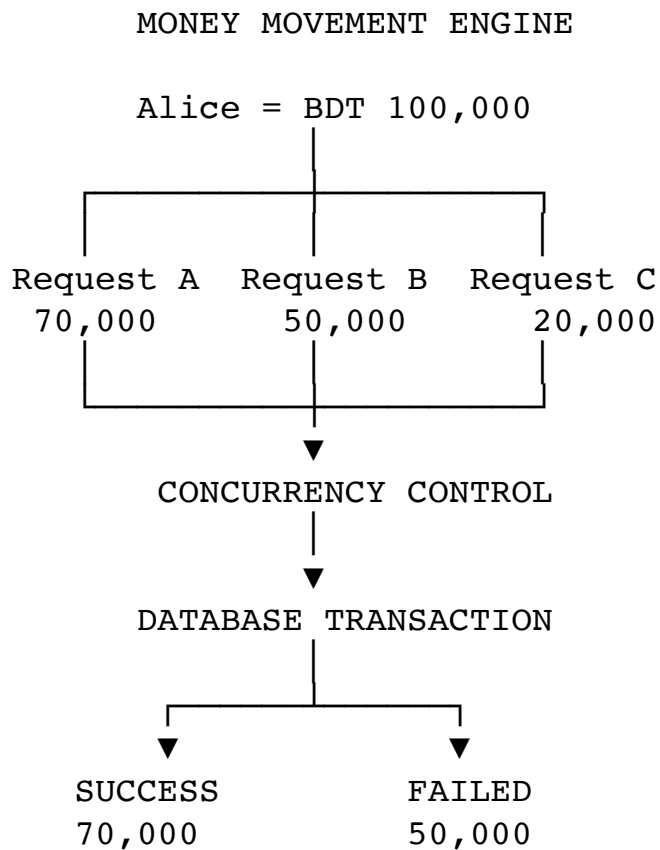
The features I would call "MUST HAVE"

If your hackathon time becomes tight, these are your **P0**:

1. User registration
2. Automatic BDT 100,000 funding
3. Login/JWT
4. Wallet/balance
5. User search
6. Send money
7. Request money
8. Accept/decline request
9. Transaction history
10. Atomic transactions
11. Row-level locking/concurrency control
12. Idempotency
13. Validation/insufficient-balance protection
14. Transaction/ledger records
15. Meaningful transaction statuses

16. Password hashing
17. Authorization
18. Rate limiting
19. Rollback/failure handling
20. Concurrent transaction tests

And your "wow" demo:



Then show:

Final balance = 30,000

and prove that the system **never created/spent money incorrectly**.

That is the part I'd make the centerpiece of the whole project.